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Backend & Data Architecture | Postgres Schema Design, JSONB Modelling, Row-Level Security | Open to remote

Database Project (Design Was Mine, End to End)

Hermes – Postgres data layer for a job-application and document dashboard | github.com/ZeppyClown/email_dashboard
Jul 2026 – Present

- Designed and own the relational schema: **content-hash text primary keys** so re-ingesting the same source row is idempotent rather than duplicating it, **CHECK**-constrained status and category enums, indexes on the four filtered columns, and a **plpgsql** trigger maintaining **updated_at**
- Split each record deliberately between real columns and one **JSONB** field: queried and constrained facts are typed columns, while per-category free-form detail is JSONB, so adding a field costs no migration and no table-per-category sprawl
- Enforced **role-scoped access in the database, not the client: row-level security** with public read, updates gated on a JWT email claim rather than **auth.role()** (which any signed-in stranger would pass), INSERT/DELETE revoked from browser roles, and **column-level UPDATE grants** limiting clients to two of sixteen columns
- Wrote the schema as its own **data dictionary**: each column carries why it exists and which alternative was rejected, and every statement is idempotent (**add column if not exists**, constraint drop-and-recreate) so the file re-runs safely against a database created earlier
- Built the read path in **plain JavaScript** over the **PostgREST** API with a three-layer fallback (live table, then manifest, then embedded snapshot), plus Python sync jobs writing through the service-role key so the browser never holds a write credential

Work Experience

DBS Bank

Singapore

Data Engineer Intern

May 2023 – May 2024

- Built a **SQL**/Grafana monitoring layer over a production pipeline running **3,000+ jobs a day**, modelling runtime and memory per pipeline version so bottlenecks could be compared across versions; teams acted on it for **20% RAM** and **15%** runtime gains
- Wrote a **REST client** for the team's **Elasticsearch** store and merged it into the shared internal Python library, giving production models real-time feature retrieval through common infrastructure for the first time

SUSS Digital Solutions & AI

Singapore

AI & Software Engineer Intern

May 2026 – Aug 2026

- Built an invoice processing pipeline extracting vendor, amount, department and billing period from PDFs, running the model **on-premises** specifically so financial documents never left the network; cut a 10-minute manual workflow to one-click review across **50+ invoices**
- Built an offline **regression harness** replaying every stored report through the scoring rules in about a second at zero API cost, then injected deliberate faults to prove each rule actually fails when violated rather than passing silently
- Caught that the headline accuracy figure had been measured on examples used to ground the judge, and re-reported it on **held-out** data, in a team codebase where the contaminated number was already in circulation

Yap Motor

Singapore

Machine Learning Intern

Dec 2025 – Jan 2026

- Shipped the company's first production AI system, a retrieval service over 500+ pages of multilingual repair manuals used daily by **24 mechanics**, still running **6 months** later; reached **90% retrieval precision** across 8 diagnostic categories
- Designed its metadata model (make, model, year parsed from free text into typed filters) and fixed two production failures traced to schema validation and case-sensitive metadata matching

Additional Projects

Music teaching platform – Firestore backend

2022

- Designed a **6-collection** schema (users, lessons, homework, notifications, chat rooms, messages) with linked lesson-to-homework records, and implemented **role-based access control** giving teacher, student and parent accounts different visibility over the same data

Drum Hub – offline sheet-music reader, published model under MIT | hf.co/Icyautumn/drum-omr-mobilenetv3

2026

Education

Nanyang Technological University, B.Comp (Computer Science), Turing AI Scholars

Aug 2024 – Dec 2028

Third year from August 2026. Coursework: Databases, Data Structures & Algorithms, Software Engineering, Machine Learning

Temasek Polytechnic, Diploma in Information Technology

2021 – 2024

Technical Skills

Databases: PostgreSQL (schema design, JSONB, indexes, CHECK constraints, triggers, views, row-level security), SQL, SQLite, Firestore, Elasticsearch, ChromaDB

Backend: Python, FastAPI, REST API design, PostgREST, Supabase, Firebase Cloud Functions, service-role vs client credential separation

Languages: Python, SQL, TypeScript, JavaScript (plain ES modules, no framework), Dart

Data & Docs: Data dictionaries, idempotent DDL, migration-safe column addition, Git, Grafana, LLM APIs and on-premises inference